ORIUS Flagship IEEE Draft: Technical Appendix, Benchmark Appendix, and Editorial Review Appendix

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I. Appendix Scope

This appendix accompanies the IEEE flagship ORIUS draft and deliberately separates technical depth from the main double-column narrative. It gathers theorem surfaces, protocol cards, benchmark-corpus material, and the editorial review appendix without creating a second source of truth beside the dissertation. The appendix reuses the same canonical monograph and generated review surfaces that govern the thesis-facing package.

II. Theory Surface Register

The IEEE appendix keeps the theorem material in register form rather than reproducing the full monograph proof chain. This makes the draft reviewable while preserving the mapping between the paper's claims, the runtime contract, and the strongest current artifact surfaces.

III. Domain Protocol Cards

The main paper compresses the six domain rows into one shared narrative. This appendix records the protocol-card view used during revision so that each row remains tied to one problem statement, one safety object, and one closure boundary.

TABLE I: Compact theory surface used by the flagship IEEE draft.

Claim family	Runtime interpretation	Strongest current row	Boundary in the IEEE draft
OASG existence	Degraded observation can make an observed-safe action true-state unsafe	Battery witness	Treated as structural across domains, but most deeply closed in the witness row.
Monotone inflation	Lower reliability widens the observation-consistent state set and tightens admissible actions	Battery witness; bounded transfer elsewhere	Claimed as a contract property, not as equal theorem depth everywhere.
Repair safety	Repaired or fallback actions reduce or bound true-state violations under sound uncertainty and repair maps	Battery plus defended bounded rows	Empirical strength varies by domain maturity.
Governed release	Certificate integrity and lifecycle transitions are part of the scientific safety object	Battery, AV, industrial, healthcare	Navigation and aerospace remain gated on full closure.

TABLE II: Protocol-card summary for the six ORIUS domain rows.

Domain	Canonical safety object	Repair mode	Fallback mode	Current closure note
Battery	State-of-charge and dispatch envelope	One-dimensional projection / clipping	Safe hold	Witness row with deepest runtime and theorem closure.
Autonomous vehicles	TTC and entry-barrier preservation	Bounded longitudinal repair	Full brake / bounded brake	Defended bounded row under the current TTC entry-barrier contract.
Industrial	Operating-envelope preservation	One-dimensional projection	Power cap	Defended bounded row for the current plant family.
Healthcare	Vital-alert and intervention envelope	Bounded intervention projection	Max alert	Defended bounded row for the current monitoring contract.
Navigation	Corridor / obstacle feasibility	Vector projection	Hold position	Gated pending real-data defended replay chain.
Aerospace	Flight-energy or envelope preservation	Bounded projection	Envelope hold	Gated pending real multi-flight runtime validation.

IV. Five-Reviewer Editorial Audit, Rebuttal, and Revision Traceability

This appendix records a simulated five-reviewer R1-style evaluation of the ORIUS monograph. The point is not artificial volume; it is to expose the manuscript to the five lenses most likely to challenge a universal physical-AI safety thesis: controls, uncertainty, formal verification, systems runtime engineering, and cross-domain thesis coherence. The tables in this appendix are editorial audit aids for revision traceability; they are not part of the scientific evidence surface.

TABLE III: Near-final reviewer scorecard summary for the ORIUS monograph.

Reviewer	Score	Primary concern
Formal Safety / Control Theory	A-	Equal-domain rhetoric must remain subordinate to the governed parity gate.
ML Uncertainty / Calibration	B+	Coverage and calibration language must stay bounded by the actual replay evidence.
CPS / Systems and Runtime Architecture	A	Runtime and governance claims must remain concrete and artifact-backed across domains.
Physical-AI Deployment and Domain Safety	B+	Navigation and aerospace still block any equal-peer deployment reading.
R1 Dissertation Committee Reader	A	The book is structurally strong, but evidence asymmetry must remain explicit everywhere.

A. Reviewer 1: Controls and Runtime Assurance

Summary judgment: The monograph is scientifically credible as a universal runtime-safety architecture, but equal-domain universality remains gated because navigation and aerospace do not yet clear the same defended closure surface as the stronger rows.

Scorecard verdict: Strong R1 thesis; equal-domain flagship claim still gated by navigation and aerospace.

Strongest contributions

- The degraded-observation hazard is identified as a plant-agnostic safety object rather than a battery-specific modeling issue.
- The theorem bridge is explicit about what is structural, what is bounded, and what must still be validated by replay or artifact.
- The parity matrix keeps the strongest claim subordinate to governed evidence rather than rhetorical flattening.

Major weaknesses

- Battery still carries the deepest theorem-to-artifact lineage, so the equal-domain narrative must remain carefully bounded.
- The formal surface remains much stronger for some domains than others.

Unsupported claims

- Any statement implying equal theorem-grade closure across all six domains.
- Any statement implying universal hardware or regulatory readiness.

Required experiments

- Real-data navigation closure under the universal replay contract.
- A stronger aerospace task with material post-repair improvement on the governing safety object.

Required writing changes

- Keep the theorem/evidence boundary stable in the theory bridge, synthesis, and conclusion chapters.
- State the parity gate once and reuse it consistently.

Decision recommendation: Weak accept for thesis submission; reject any equal-domain flagship claim until the gated rows are closed.

B. Reviewer 2: Statistical ML and Conformal Inference

Summary judgment: The uncertainty layer is valuable because it treats reliability-conditioned uncertainty as a runtime safety object, but the book must remain precise about where coverage language is formal and where it is empirical or conservative.

Scorecard verdict: The uncertainty layer is credible; the monograph must still stop short of equal-domain statistical closure.

Strongest contributions

- The monograph ties conformal and calibration ideas to runtime action legality rather than to prediction quality alone.
- The universal benchmark schema makes true-state violation and observed-state satisfaction separable across domains.
- The non-i.i.d. and distribution-shift caveats are visible instead of hidden.

Major weaknesses

- The weaker rows do not yet support equal-domain statistical closure.
- Coverage-style language can still be overread if chapter summaries are not carefully bounded.

Unsupported claims

- Universal statistical calibration under arbitrary shift.
- Formal coverage guarantees for domains whose replay surface remains portability-only or experimental.

Required experiments

- Fault-mode and domain-specific coverage diagnostics for every defended row.
- Additional uncertainty summaries for navigation and aerospace once their evidence surfaces are upgraded.

Required writing changes

- Differentiate bounded calibration claims from conservative widening heuristics.
- Keep uncertainty claims tied to tracked replay artifacts.

Decision recommendation: Accept for a bounded universal-safety monograph; major revision for any stronger cross-domain statistical claim.

C. Reviewer 3: Systems and Runtime Engineering

Summary judgment: The systems contribution is strong because runtime behavior, fallback, certificates, and audit continuity are first-class objects in the main manuscript rather than hidden implementation details.

Scorecard verdict: Systems contribution is strong enough for serious review once the parity matrix stays central.

Strongest contributions

- CertOS is integrated into the scientific argument rather than treated as release tooling.
- The replay and benchmark contract is specific enough to support cross-domain inspection.
- The backend artifact path and governed publication surface keep the evidence path auditable.

Major weaknesses

- Runtime and fallback evidence are still deepest in the witness and strongest defended rows.
- Cross-domain composition and lifecycle coverage remain selective.

Unsupported claims

- Full deployment readiness in every domain.
- Uniform multi-agent or shared-constraint closure across all rows.

Required experiments

- Broader runtime-budget and lifecycle traces for the weaker domains.
- Expanded cross-domain composition scenarios where the adapter contract actually supports them.

Required writing changes

- Keep latency, fallback, certificate validity, and audit continuity in the mainline claims.
- Avoid relegating critical runtime semantics to appendices only.

Decision recommendation: Strong accept as a systems-and-governance thesis; bounded revision required before broader deployment claims.

D. Reviewer 4: Cross-Domain Physical AI and Deployment

Summary judgment: ORIUS has credible societal potential as a fundamental safety layer for physical AI, but that potential only reads as serious when the monograph keeps weaker domains explicitly gated instead of rhetorically promoted.

Scorecard verdict: The physical-AI safety layer claim is compelling when bounded deployment language is preserved.

Strongest contributions

- The book treats battery, AV, industrial, healthcare, navigation, and aerospace as first-class domain chapters rather than as add-on examples.
- The parity gate prevents the deployment narrative from outrunning the artifact surface.
- The domain chapters share one template, which makes the universality claim inspectable.

Major weaknesses

- Navigation and aerospace remain below equal-peer maturity.
- Some readers will still look for broader field closure than the current evidence supports.

Unsupported claims

- Equal defended deployment credibility across every domain.
- Aerospace or navigation as mature peers to the witness and strongest defended rows.

Required experiments

- A defended real-data navigation row.
- A non-placeholder aerospace benchmark with a stronger safety object and post-repair gain.

Required writing changes

- Retain explicit domain non-claims in every chapter.
- Keep deployment language bounded by the parity matrix and artifact register.

Decision recommendation: Accept as a universal physical-AI safety architecture; major revision for any flat equal-domain deployment claim.

E. Reviewer 5: R1 Dissertation and Monograph Coherence

Summary judgment: The manuscript now reads like a real R1-style monograph because it has one hazard, one architecture, one theorem bridge, six domain chapters, and one parity gate. The remaining gap is evidence symmetry, not book structure.

Scorecard verdict: Book quality is near submission-ready; the remaining gap is evidence parity, not structure.

Strongest contributions

- The main body no longer depends on stitched paper-lineage framing.
- The universal-first narrative is visible from the abstract through the conclusion.
- The appendices provide bibliography, review, protocol, crosswalk, and artifact traceability at book scale.

Major weaknesses

- Legacy repo surfaces can still contradict the monograph if they are not treated as archival support only.
- The equal-domain ambition is stronger than the current parity evidence.

Unsupported claims

- A claim that the monograph has already earned equal-domain closure.
- Any suggestion that weaker rows are present only for symmetry rather than bounded scientific use.

Required experiments

- Only the parity-closing experiments needed for navigation and aerospace if equal-domain promotion remains the target.

Required writing changes

- Keep the main body universal-first and the weaker rows honest.
- Preserve the explicit distinction between architecture universality and evidence parity in the introduction, synthesis, and conclusion.

Decision recommendation: Accept as a thesis-length monograph with bounded universal claims; defer stronger flagship universality language until parity is actually earned.

F. Consolidated Editorial Findings

TABLE IV: Consolidated editorial findings across the five reviewer lenses.

Gap	Where it appears	Reviewer pressure	Revision action
Evidence asymmetry across domains	Parity matrix, cross-domain synthesis, and domain status language	Controls, deployment, and committee readers all insist that architecture universality must not be confused with equal-domain closure	Keep battery as witness, AV plus industrial and healthcare as defended bounded rows, navigation as shadow-synthetic, and aerospace as experimental until stronger reruns exist.
Coverage and calibration ambiguity	Related work, uncertainty discussion, and benchmark interpretation	Statistical-ML review demands a clear line between formal calibration claims and conservative widening heuristics	State where coverage language is theorem-backed, where it is replay-backed, and where it is only heuristic.
Theorem / replay / implementation drift	Theory bridge, synthesis, and conclusion	Formal review demands an explicit claim boundary that is reused verbatim	Tie every strong claim to a theorem object, tracked artifact, or explicit bounded empirical result.
Runtime and audit visibility	Governance, latency, and fallback chapters	Systems review requires runtime budgets, certificate lifecycle, and audit continuity to stay in the main narrative	Keep governance and fallback semantics first-class in the architecture and synthesis chapters.
Legacy framing leakage	Non-canonical repo docs and archival artifacts	Committee and deployment readers will downgrade the book if older battery-origin or stitched-paper language leaks into the main story	Keep the canonical monograph universal-first and treat archival paper-lineage artifacts as support only.

Family	Rows
Safety filters and constrained control	18
General physical AI and systems	17
Runtime assurance and monitoring	14
Conformal and calibration	12
Degradation, anomaly, and CPS security	10
Energy and storage	9
Robotics and autonomy	8
Industrial automation	5
Healthcare monitoring	4
Navigation and aerospace	3

TABLE VI: Family distribution for the 100-paper benchmark corpus used to pressure-test the ORIUS flagship draft.

G. Author Rebuttal

ORIUS does not need to claim equal empirical maturity in every domain to justify the monograph. It needs to justify one universal runtime safety layer and then state, with governed discipline, how each domain currently maps onto that contract.

- **To the controls reviewer**: the monograph keeps the parity gate reader-visible and does not blur architecture with equal closure.
- **To the statistical-ML reviewer**: calibration language is bounded to the evidence surface and never upgraded into universal statistical guarantees by prose.
- **To the systems reviewer**: runtime behavior, fallback, certificate validity, and audit continuity remain part of the main scientific contribution rather than appendix-only material.
- **To the deployment reviewer**: weaker rows remain explicitly gated so the physical-AI safety-layer claim does not borrow credibility from unclosed domains.
- **To the committee reviewer**: the book keeps one universal narrative spine from hazard to parity gate and uses the appendices only to deepen, not replace, the main argument.

H. Revision Traceability

TABLE V: Revision traceability matrix from reviewer critique to manuscript action.

Reviewer	Critique	Revision action / file target
Controls and runtime assurance	Do not blur universal architecture with equal-domain parity.	Keep the parity gate central in `monograph/ch14_cross_domain_synthesis.tex` and align all domain labels to the tracked parity matrix.
Statistical ML / conformal	Separate formal calibration from conservative widening.	Keep claim-boundary language explicit in `monograph/ch02_oasg_claim_boundary.tex` and the term register.
Systems and runtime engineering	Keep governance, fallback, latency, and certificate lifecycle visible.	Preserve runtime and governance emphasis in `monograph/ch07_system_benchmark_governance.tex` and the supporting CertOS chapters.
Cross-domain deployment	Keep navigation and aerospace gated until stronger evidence exists.	Maintain explicit non-claims and promotion obligations in the domain chapters and protocol-card appendix.
R1 monograph coherence	Preserve one universal-first narrative spine and treat archival language as non-canonical.	Keep `paper/paper.tex`, the monograph chapters, and the authoring guide synchronized around the same universal-first structure.

V. Benchmark Corpus Summary

a) Writing-gap synthesis.: The benchmark corpus pushes the ORIUS flagship draft toward five consistent writing upgrades: lead with the field-level problem instead of artifact logistics; state one dominant technical move early; compress

evidence to a few decisive surfaces; keep limitations adjacent to the main claim; and make the conclusion argue for a field object rather than merely restating chapter coverage.

VI. Top-Tier Benchmark Corpus

This appendix benchmarks ORIUS against a broad top-tier research corpus spanning conformal inference, runtime assurance, safe control, anomaly and drift detection, cyber-physical systems, and domain-specific safety literature. The corpus is used as a writing and positioning benchmark rather than as a numerical leaderboard.

TABLE VII: Broad benchmark corpus used to pressure-test the flagship ORIUS draft.

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2023	top-tier broad	Safety filters and constrained control	Data-Driven Safety Filters: A Survey and Perspective	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2022	top-tier broad	Safety filters and constrained control	Battery Storage Operation Under Uncertainty: A Distributionally Robust Perspective	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2021	top-tier broad	Safety filters and constrained control	Distribution-Free, Risk-Controlling Prediction Sets	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2021	top-tier broad	Safety filters and constrained control	Safe Reinforcement Learning: A Control-Theoretic Perspective	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2021	top-tier broad	Safety filters and constrained control	Safety-Critical Control for Autonomous Systems	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2019	top-tier broad	Safety filters and constrained control	A General Safety Framework for Learning-Based Control in Uncertain Robotic Systems	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2019	top-tier broad	Safety filters and constrained control	A Survey on Safe Reinforcement Learning	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2019	top-tier broad	Safety filters and constrained control	A Systems and Control Perspective of CPS Security	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2019	top-tier broad	Safety filters and constrained control	End-to-End Safe Reinforcement Learning Through Barrier Functions for Safety-Critical Continuous Control Tasks	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2019	top-tier broad	Safety filters and constrained control	Human Compatible: Artificial Intelligence and the Problem of Control	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2018	top-tier broad	Safety filters and constrained control	Learning-Based Model Predictive Control for Safe Exploration	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2018	top-tier broad	Safety filters and constrained control	Planning and Decision-Making for Autonomous Vehicles	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2017	top-tier broad	Safety filters and constrained control	Safety Barrier Certificates for Collisions-Free Multirobot Systems	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2016	top-tier broad	Safety filters and constrained control	Exponential Control Barrier Functions for Enforcing High Relative-Degree Safety-Critical Constraints	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2016	top-tier broad	Safety filters and constrained control	Survey and New Directions for Physics-Based Attack Detection in Control Systems	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2015	top-tier broad	Safety filters and constrained control	Robustness of Control Barrier Functions for Safety Critical Control	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2013	top-tier broad	Safety filters and constrained control	Attack Detection and Identification in Cyber-Physical Systems	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2013	top-tier broad	Safety filters and constrained control	Model Predictive Control	control-theoretic with compact proofs	must expose one reusable cross-domain contract instead of plant-local mathematics only
2024	top-tier broad	General physical AI and systems	Physical AI Systems: Architectures, Risks, and Safety Layers	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2023	top-tier broad	General physical AI and systems	Are Transformers Effective for Time Series Forecasting?	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2023	top-tier broad	General physical AI and systems	Artificial Intelligence Risk Management Framework (AI RMF 1.0)	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2023	top-tier broad	General physical AI and systems	TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2022	top-tier broad	General physical AI and systems	Deep Neural Networks and Tabular Data: A Survey	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2022	top-tier broad	General physical AI and systems	FEDformer: Frequency Enhanced Decomposed Transformer for Long-Term Series Forecasting	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2021	top-tier broad	General physical AI and systems	An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2021	top-tier broad	General physical AI and systems	Argoverse 2: Next Generation Datasets for Self-Driving Perception and Forecasting	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2021	top-tier broad	General physical AI and systems	Cautious Adaptation for Reinforcement Learning under Safety Constraints	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2021	top-tier broad	General physical AI and systems	Improving Reproducibility in Machine Learning Research	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2021	top-tier broad	General physical AI and systems	Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2021	top-tier broad	General physical AI and systems	Self-Driving Cars: A Survey	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2020	top-tier broad	General physical AI and systems	DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2019	top-tier broad	General physical AI and systems	Evaluating Model Calibration in Classification	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2019	top-tier broad	General physical AI and systems	Measuring Calibration in Deep Learning	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2019	top-tier broad	General physical AI and systems	Software Engineering for Machine Learning: A Case Study	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone
2022	top-tier broad	Runtime assurance and monitoring	Industrial Process Monitoring and Fault Diagnosis: A Survey	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2020	top-tier broad	Runtime assurance and monitoring	Runtime Assurance for Safe Learning-Enabled Aerospace Systems	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2020	top-tier broad	Runtime assurance and monitoring	Shielding Techniques for Safe Reinforcement Learning	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2019	top-tier broad	Runtime assurance and monitoring	SOTER: A Runtime Assurance Framework for Programming Safe Robotics Systems	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2018	top-tier broad	Runtime assurance and monitoring	Introduction to Runtime Verification	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2018	top-tier broad	Runtime assurance and monitoring	Safe Reinforcement Learning via Shielding	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2014	top-tier broad	Runtime assurance and monitoring	A Review on Basic Data-Driven Approaches for Industrial Process Monitoring	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2012	top-tier broad	Runtime assurance and monitoring	Assurance Cases for Runtime Assurance Frameworks	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2012	top-tier broad	Runtime assurance and monitoring	Gaussian Processes for Personalized E-Health Monitoring	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2011	top-tier broad	Runtime assurance and monitoring	Sandboxing Controllers for Cyber-Physical Systems	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2005	top-tier broad	Runtime assurance and monitoring	LOLA: Runtime Monitoring of Synchronous Systems	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2004	top-tier broad	Runtime assurance and monitoring	An Overview of the Runtime Verification Tool Java PathExplorer	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2004	top-tier broad	Runtime assurance and monitoring	Efficient Decentralized Monitoring of Safety in Distributed Systems	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
1998	top-tier broad	Runtime assurance and monitoring	The Simplex Architecture for Safe Online Control System Upgrades	systems-first with mechanism diagrams	must bind supervision to degraded observation and benchmark semantics
2023	top-tier broad	Conformal and calibration	Conformal Prediction Beyond Exchangeability	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2021	top-tier broad	Conformal and calibration	A Comparison of Some Conformal Quantile Regression Methods	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2021	top-tier broad	Conformal and calibration	Calibrated Multiple-Output Quantile Regression with Conformal Prediction	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2021	top-tier broad	Conformal and calibration	Conformal Prediction Interval for Dynamic Time-Series	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2021	top-tier broad	Conformal and calibration	Distributional Conformal Prediction	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2021	top-tier broad	Conformal and calibration	Online Conformal Prediction with Decaying Step Sizes	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2008	top-tier broad	Conformal and calibration	A Tutorial on Conformal Prediction	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2005	top-tier broad	Conformal and calibration	Conformal Predictors and Confidence Machines	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2023	legacy carry-forward	Conformal and calibration	Conformal prediction beyond exchangeability	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2023	legacy carry-forward	Conformal and calibration	Conformal prediction: A gentle introduction	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2023	legacy carry-forward	Conformal and calibration	Testing for outliers with conformal p-values	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2022	legacy carry-forward	Conformal and calibration	Adaptive conformal predictions for time series	theorem-forward with compact experiments	must connect coverage to repaired actuation and certificates
2019	top-tier broad	Degradation, anomaly, and CPS security	Survey of Intrusion Detection Systems: Techniques, Datasets and Challenges	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2015	top-tier broad	Degradation, anomaly, and CPS security	Principles of Cyber-Physical Systems	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2015	top-tier broad	Degradation, anomaly, and CPS security	The Past, Present and Future of Cyber-Physical Systems: A Focus on Models	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2012	top-tier broad	Degradation, anomaly, and CPS security	Modeling Cyber-Physical Systems	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2010	top-tier broad	Degradation, anomaly, and CPS security	Cyber-Physical Systems: The Next Computing Revolution	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2008	top-tier broad	Degradation, anomaly, and CPS security	Cyber Physical Systems: Design Challenges	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2001	top-tier broad	Degradation, anomaly, and CPS security	Fault Detection and Diagnosis in Industrial Systems	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2018	legacy carry-forward	Degradation, anomaly, and CPS security	A survey of physics-based attack detection in cyber-physical systems	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2010	legacy carry-forward	Degradation, anomaly, and CPS security	Medical cyber physical systems	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2009	legacy carry-forward	Degradation, anomaly, and CPS security	Anomaly detection: A survey	taxonomy or threat-model driven	must connect alarms to action semantics and repaired releases
2020	top-tier broad	Energy and storage	A Comprehensive Review on Grid-Connected Battery Energy Storage Systems	application-heavy with cost and safety tables	must generalize beyond the witness plant
2020	top-tier broad	Energy and storage	A Review of Battery Energy Storage System Safety and Reliability	application-heavy with cost and safety tables	must generalize beyond the witness plant
2020	top-tier broad	Energy and storage	Modeling of Lithium-Ion Battery Degradation for Cell Life Assessment	application-heavy with cost and safety tables	must generalize beyond the witness plant

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2020	top-tier broad	Energy and storage	Risk-Averse Battery Dispatch Under Forecast Error	application-heavy with cost and safety tables	must generalize beyond the witness plant
2019	top-tier broad	Energy and storage	Data-Driven Prediction of Battery Cycle Life Before Capacity Degradation	application-heavy with cost and safety tables	must generalize beyond the witness plant
2018	top-tier broad	Energy and storage	Energy Storage Dispatch in Energy Markets: A Review	application-heavy with cost and safety tables	must generalize beyond the witness plant
2016	top-tier broad	Energy and storage	Battery Management Systems, Volume III: Battery State Estimation	application-heavy with cost and safety tables	must generalize beyond the witness plant
2016	top-tier broad	Energy and storage	Critical Review of State of Health Estimation Methods of Li-Ion Batteries for Real Applications	application-heavy with cost and safety tables	must generalize beyond the witness plant
2015	top-tier broad	Energy and storage	Battery Management Systems, Volume II: Equivalent-Circuit Methods	application-heavy with cost and safety tables	must generalize beyond the witness plant
2023	top-tier broad	Robotics and autonomy	UL 4600: Standard for Safety for the Evaluation of Autonomous Products	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2021	top-tier broad	Robotics and autonomy	Large Scale Interactive Motion Forecasting for Autonomous Driving: The Waymo Open Motion Dataset	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2020	top-tier broad	Robotics and autonomy	nuScenes: A Multimodal Dataset for Autonomous Driving	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2018	top-tier broad	Robotics and autonomy	ISO 26262 Road Vehicles – Functional Safety	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2017	top-tier broad	Robotics and autonomy	Autonomous Vehicle Safety: An Interdisciplinary Challenge	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2017	top-tier broad	Robotics and autonomy	Examining Accident Reports Involving Autonomous Vehicles in California	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2017	top-tier broad	Robotics and autonomy	FaSTrack: A Modular Framework for Fast and Guaranteed Safe Motion Planning	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar
2017	top-tier broad	Robotics and autonomy	FaSTrack: A Modular Framework for Fast and Guaranteed Safe Motion Planning	figure-driven and benchmark-oriented	must unify autonomy rows with non-robotics CPS rows under one grammar

Year	Tier	Family	Paper	Writing lens	ORIUS gap
2021	top-tier broad	Industrial automation	Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting	operational and systems-focused	must expose typed repair, fallback, and certificate semantics
2019	top-tier broad	Industrial automation	PyTorch: An Imperative Style, High-Performance Deep Learning Library	operational and systems-focused	must expose typed repair, fallback, and certificate semantics
2017	top-tier broad	Industrial automation	A Unified Approach to Interpreting Model Predictions	operational and systems-focused	must expose typed repair, fallback, and certificate semantics
2017	top-tier broad	Industrial automation	Attention Is All You Need	operational and systems-focused	must expose typed repair, fallback, and certificate semantics
2017	top-tier broad	Industrial automation	Revisiting the Tennessee Eastman Process Simulation Benchmark for Industrial Machine Learning	operational and systems-focused	must expose typed repair, fallback, and certificate semantics
2023	top-tier broad	Healthcare monitoring	MIMIC-IV, a Freely Accessible Electronic Health Record Dataset	clinical framing with conservative claims	must connect degraded observation to runtime intervention legality
2019	top-tier broad	Healthcare monitoring	A Guide to Deep Learning in Healthcare	clinical framing with conservative claims	must connect degraded observation to runtime intervention legality
2018	top-tier broad	Healthcare monitoring	The eICU Collaborative Research Database, a Freely Available Multi-Center Database for Critical Care Research	clinical framing with conservative claims	must connect degraded observation to runtime intervention legality
2017	top-tier broad	Healthcare monitoring	Opportunities and Challenges in Developing Risk Prediction Models with Electronic Health Records Data	clinical framing with conservative claims	must connect degraded observation to runtime intervention legality
2020	top-tier broad	Navigation and aerospace	OpenSky Report 2020: Analysing In-Flight Emergency Events Using Big Data	safety-case oriented with explicit limits	must clear defended real-data replay before equal-domain rhetoric
2015	top-tier broad	Navigation and aerospace	Run-Time Assurance for Complex Systems	safety-case oriented with explicit limits	must clear defended real-data replay before equal-domain rhetoric
2011	legacy carry-forward	Navigation and aerospace	DO-178C: Software Considerations in Airborne Systems and Equipment Certification	safety-case oriented with explicit limits	must clear defended real-data replay before equal-domain rhetoric
2019	top-tier broad	General physical AI and systems	Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead	broad positioning and synthesis	must be tied to a concrete runtime contract rather than vision alone

VII. Editorial Ledgers

The flagship IEEE draft is accompanied by a private claim-delta ledger and a revision ledger. They are not part of the scientific evidence surface, but they are useful for editing discipline because they distinguish between claims already supported by tracked repo artifacts, claims supported primarily by literature, and claims that still require closure work before an artifact-strict submission.

TABLE VIII: Support-status tags used in the private editorial ledgers.

Tag	Meaning
`current_repo_supported`	Claim aligns with tracked ORIUS artifacts or governed release surfaces.
`supported_by_primary_literature_only`	Claim is well grounded in primary sources but not yet fully closed by local artifacts.
`visionary_requires_closure_work`	Claim is directionally important but still ahead of present closure evidence.

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